Niels Bjørn-Andersen (ed.)

CASES ON IT LEADERSHIP

CIO Challenges for Innovation and Keeping the Lights on

1. edition

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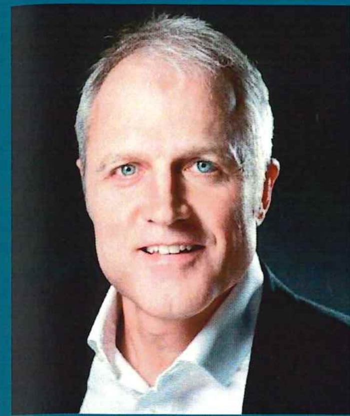

RENÉ MUNK-NISSEN

1.a The Challenge of Technological Debt – Coop

Jonas Hedman and Niels Bjørn-Andersen

Introduction

It was a lovely autumn day in 2015 when René Munk-Nissen drove to the Coop Head Office in Albertslund, but he was not enjoying the weather. Since October 1, 2012, he had been the CIO of the largest retail group in Denmark, where he had been headhunted to create a high performance IT function that would deliver value to the organization 24/7.

He had accomplished this task. 'There was no silver bullet; basically it was a lot of really hard work,' he thought to himself, but he was proud of his team. They had created a much higher performing IT function engaging modern principles for IT development and governance, but the technological legacy of the old systems still needed IT investments of more than 2 billion DKK. But that was simply hard work, for which plans were already in place. The real challenge and problem were that it was difficult for the management board to comprehend the magnitude and really understand the sense-of-urgency for immediately confronting the critical state of the IT infrastructure.

For 2015 and 2016, top management had asked him to continue running with the old infrastructure even though it was becoming more and more complex and difficult to maintain a stable IT operation, just 'keeping the lights on.' He could not guarantee that the mainframe-based infrastructure could secure business continuity, and potentially could even damage it. That would then introduce substantial operational costs or inhibitors to the core business.

Today, he and the management board had been summoned to discuss the IT budget for 2016 in the context of Coops forecasted profitability. The message was gloomy. In line with other HQ functions, IT also had to manage on an even more constrained budget than in 2015. In other words, Munk-Nissen did not get a firm commitment to what he believed was absolutely necessary, i.e., to commence the investment of renovating the IT infrastructure. Accordingly, he found it increasingly difficult professionally to take responsibility without what he felt was unsatisfactory disclaimers about high risk. On December 2, 2015, René Munk-Nissen drove home from the Head Office for the last time.

CIO René Munk-Nissen

René Munk-Nissen had come to Coop with the best credentials. His formal qualifications included a degree as software engineer and a BA in IT strategy and Management, which he had supplemented with an MBA (Ashridge) and courses in Strategy and Change Management (IMD).

In total, he had more than 25 years of experience in IT. He started as an application developer for Egmont and Codan, but moved quickly into five management positions. He had been CIO or held similar positions in no less than different organizations in five different industries over a period of 16 years in national as well as international positions. He had extensive work experience, including senior positions in Fona (Project Manager), Royal Unibrew (Group CIO), Carl Bro Group (CIO), Actavis Group (VP for global ERP and IT WEMEA) and DSB (SVP and CIO), before he took up the position as Senior Vice President and CIO for Coop, Denmark.

Coop and the competitive landscape of supermarkets in Denmark

Following examples from the UK, the history of Coop went back to the late 19th century when the Cooperative movement started taking shape in Denmark and other Nordic countries. Coop's first store opened in Denmark in 1866. Currently holding. 1,5 million owners, the Cooperative legacy and history differentiated Coop from all other grocery chains.

In 2015, Coop was the market leader in a strongly competitive cut-throat market for supermarket goods. Currently, the biggest market shares belong to Coop (approximately 37%) and Dansk Supermarked (approximately 34%), and both of these groups have a full range of shops from large supermarkets to smaller corner stores, plus discount stores. What is especially challenging to the market is the growth in discount stores, such as Aldi, Lidl, Kiwi, and Rema 1000. Some of these are opening more shops in spite of a much too high overall coverage of retail shops in Denmark, compared to other countries, and they are increasing their market share despite a very low profit or some are even suffering financial losses. Coop has been able to increase its market share in comparison to its biggest competitor – Dansk Supermarked.

In 2015, Coop was the biggest employer of any organization in Denmark with 38,000 employees, a turnover of approximately 45 billion DKK, and about 1,200 stores across the country. Coop had issued some 1,500,000 customer loyalty cards, and it was serving more than 10% of the Danish population on a daily basis.

However, Coop had been struggling for many years to deliver a reasonably profitable result illustrated by the fact that Coop was only delivering a profit margin of approximately 1% in the industry. It was a misconception that Coop was a 'not-for-profit' organization. It was never meant to be that, but it had been extremely difficult to make substantial profit. Accordingly, there had been a high turnover in the CEO position and in many top management positions in Coop over the last ten years.

In 2015, Coop Denmark had five supermarket chains – Kvickly, SuperBrugsen, Dagli' Brugsen, Fakta, and Irma – with 1,200 physical shops. The Fakta stores were divided into two groups, normal Fakta stores and the smaller convenience stores Fakta Q that had a more limited selection of products. Coop also had two web-shops: Coop.dk and Irma.dk (see Figure 1 for the number of stores in each chain).

The majority of the stores were owned by Coop, but approximately 400 of the

stores (typically in the Dagli' Brugsen, Kvickly and SuperBrugsen chains) were owned by individual storeowners (franchises). Many of these individually owned shops were performing better than the Coop owned stores. However, the individually owned stores also presented a huge challenge when it came to major investments in Head Office, such as e.g., investments in IT. This was because it was sometimes difficult for a franchisee to see the benefits of investments in IT infrastructure or an integration/standardization of systems for several chains in order to create an overall more effective business processes and economy of scale for all in the long run.

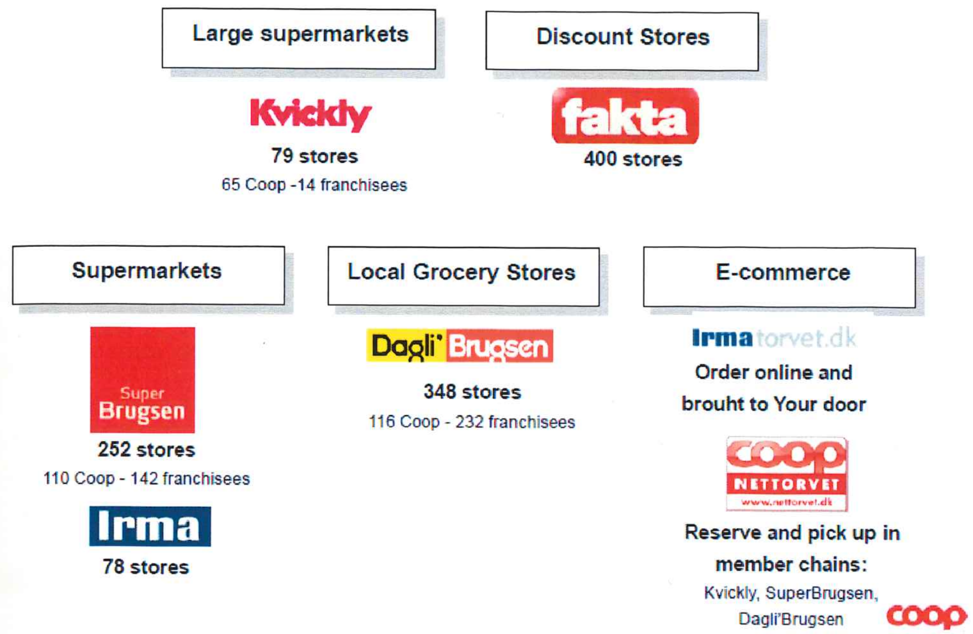

Figure 1. Number of stores for each of the five chains in February 2013

In order to serve the stores, Coop had a very complex supply chain and logistics (see Figure 2 for an overview). It involves, a 24/7 operation – with 1,800 employees, 550 trucks, and 4,300 store deliveries every day. The supply chains

are partly managed by Coop and partly out-sourced in order to have the most effective supply-chain operations. For instance, the management of fresh food terminals has been outsourced to Arla, the large diary product supplier. For more details, see Appendix 1.

In order to deliver superior value to customers, Coop needs to consolidate the supply chain. For instance, the head of non-food and the head of food are negotiating and buying products on behalf of all the chains. Since this is a low-margin industry, efficiency in supply chain and logistics are key for Coops' success.

The two Internet shops, Coop.dk and Irma.dk, have a high number of visitors, but it is a challenge to make a profit through online sales at this stage. The logistic cost of the 'last mile to the consumer' is high. Unfortunately, delivery costs seem to be higher than the average consumer is willing to pay, and prices in web-shops cannot be higher than in physical stores. Furthermore, there is limited – if any – integration between the Internet shops. They even use different IT infrastructure and point of sales (POS) systems.

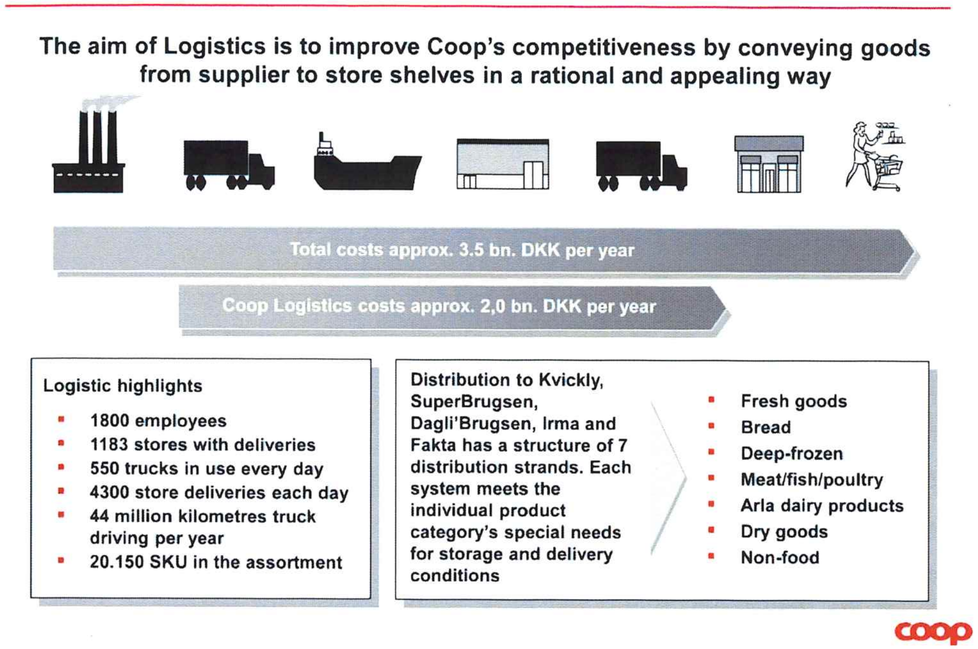

Figure 2. The logistics and supply chain of Coop

Historically and in line with the industry reality, Coop's competitive strategy had been focused on two parameters: increase the number of stores as well as offer competitive and discounted prices. However, CIO Munk-Nissen saw this strong focus on price in a market segment with over-capacity (high cost structure) as a dead man's game, since there were already too many stores and too many square meters of supermarket shops in Denmark. Accordingly, he would have liked to focus his attention on turning IT into a competitive and strategic advantage for Coop to provide customers with more added value.

The role of IT in Coop

When René Munk-Nissen joined COOP in October 2012, he felt that it was time for a paradigm shift in the retail industry (similar to the one that occurred in the 1980s when discount shops entered into the market) because it was very hard for any of the retailors to make a profit. With an estimated turnover of around DDK 120 billion in the supermarket retail sector, it was simply unacceptable that the sector as a whole was making less than DDK 2 billion in profit. Something had to be done.

> We need to bring value to the end customers, to get their attention, to get more customer traffic into our shops, and to increase basket sizes. What I'm looking into is to turn IT into a competitive edge for the business. ... I believe that we can certainly make a difference... It is very clear that if IT is not working it will stop the flow of milk to the shelves, and that will be very costly or even long-term destroy the company. Furthermore, I strongly believe that utilization and digitalization can be turned into a business differentiator, and hence we need to add value to the game. (Munk-Nissen 2013)

Consequently, one of the first tasks for the new CIO in 2012 and early 2013 was to review the IT function. The two main areas investigated were the IT technical landscape and the IT organization. Based on that assessment, it was possible in Q2 2013 to propose an IT strategy for Coop.

During his first six months as CIO, Munk-Nissen reengineered and trimmed the IT function to meet the demands of Coop.

> We really needed a bi-model organization. One part of the IT-function should be able to do the massive, reliable, transaction-based task, while another part of the IT-function had to be fast moving, effective, and capable of meeting the pace of the business needs without delays. (Munk-Nissen 2015)

Accordingly, the goal was to have an IT function for the future that would support the business in having stable operations, instant scalability, and integrated changeability. One of the issues was the lack of integration of Coop's various IT applications and databases. The explanation for this goes beyond just the complexity and the number of applications and databases. Historically, Coop has been a do-it-yourselves IT-shop, but this was something the new CIO of Coop wanted to turn into strategic sourcing in order to better support stability, agility, and the dynamic of the core business. One illustration of the challenges was the status of Coop's Electronic Data Interchange (EDI) critical for the supply chain and flow of goods, which basically was one of the main blood veins of the company:

> The EDI systems are more than 20 years old, and they are homemade. There is only one guy left. He is two months from retirement and in poor health, and actually, he

> has now been hospitalized. However, if EDI is not running, nothing is happening. So that is one of the critical and high risk single point of failures I'm looking into right now. (Munk-Nissen 2013)

Clearly, the previous strategy of developing in-house was not sustainable. Whatever IT there was to be built or delivered to Coop should be provided by the best solution provider – whether that be Coop itself or a professional provider from the market. As Munk-Nissen says:

> It is quite clear that IT is absolutely mandatory for Coop. We should provide the very best service, whether it is via systems developed by ourselves or we act as facilitator of IT services. We need to be an efficient it-broker to deliver the best it service and digitalization there is. (Munk-Nissen 2013)

Taking stock of the IT landscape

The main objective for Munk-Nissen was to provide IT service with a 100% reliability, low costs, high effectiveness, and high flexibility to meet the new and rapidly changing business demands. This required a highly effective IT infrastructure. However, that was hardly what René Munk-Nissen found when he first arrived. The complexity of the task was evident when one considers that Coop IT had to provide efficient operations, maintenance, and development for

- 24/7 IT service operation 365 days a year
- Helpdesk handling approximately 500,000 helpdesk requests by phone or email annually for stores, warehouses, and Head Office
- Business Intelligence: Data Warehouse (all goods – all stores – every day)
- EDI with over 900 suppliers
- Two web shops: COOP.dk and Irma.dk
- Store, salary, accounting, logistics, HR, and systems
- CRM and goods systems
- Telephone system and network
- 5,200 point of sales terminals
- 5,000 + PCs
- 2,000 + servers
- 650 applications
- 398 databases and data models
- 55,000 software programs

Furthermore, Coop was using almost any conceivable technology platform from companies such as SAP, Microsoft, Citrix, Sun, Cisco, Oracle, etc.

It was evident that COOP was struggling with a very complex IT infrastructure. The IT landscape had a high number of applications, platforms, components, and databases in an environment of old systems and legacy mainframes that were more than 20 years old. The mainframe was key to 650 software applications of which almost 100 applications had exceeded the end-of-support date and urgently needed upgrading (or accepting the business risk). To keep it running 24/7/365 demanded a lot of work, planning, and operational skills. Some of the employees specialized in the old systems were required not to go on vacation without their laptop. Even more critical, the mainframe systems were the backbones, and if the mainframe went offline, everything would come to a grinding halt.

One of the critical issues was the lack of integration of the various IT applications and databases. The explanation for this went beyond just the mere complexity and the number of applications and databases. Historically, each of the five retail chains had a certain level of autonomy, and there were several different cashier terminals (POS systems) and logistics systems to maintain.

This huge complexity also means that it was very costly to drive any type of innovation. If, e.g., the retail chains would like to introduce a food recipe for customers in such a way that the groceries for the recipe were available in all chains, it would require program development in systems for several chains, since there was no joint platform for customer interfacing systems, e.g. web- and smartphone applications. Another example, if Coop would like to introduce a voice controlled picking system in the warehouses, it had to be implemented in three different warehouse systems. For a third example, the current legacy systems or technological debt was preventing any serious development in the direction of an omni-/multichannel solution across Coops different retail formats integrating both brick-and-mortar stores and e-commerce sales channels.

In summary, running so many different application programs on so many different platforms would be both very costly to maintain and very susceptible to errors. This would not be sustainable for high business continuity and for time-to-market in an almost 24x7 business in a highly competitive market.

A major undertaking to remedy part of these challenges was to develop a new joint HR and Finance system using SAP in 2013. In spite of the fact that it was standard ERP modules, a substantial development effort had to be mounted due to different needs of the retail chains, the needs of the HQ, and the special needs of the franchise shops. The original budget of approximately DKK 250 million was substantially exceeded in terms of costs and time to implement.

Developing a high-performance IT organization

Another major challenge was the IT organization. Before Munk-Nissen joined the company in October of 2012, the IT function had some 200 employees, but there had been cost reductions and lay-offs in late 2011 and in 2012, and the IT group was now down to 140. This was not enough to manage the very complex IT environment, as well as a number of big vendor and outsourcing contracts. Accordingly, Munk-Nissen began to upgrade his IT management team by hiring a number of new people as part of an ongoing change process.

When Munk-Nissen took over, the IT organization was very traditionally divided into development and operations. Within each of these two divisions, work was very much specialized in silos, due to the many different platforms and applications.

In order to create a high-performance IT organization, major changes had to be implemented. A high-level overview of the blueprint for the future IT function is shown in Figure 3. To the left is shown the five retail chains, the two web-shops, and the major business processes. The central part of the figure still has IT development and IT services, but in addition, two new units were launched responsible for strategy, governance, and business partnering. This included all management functions dealing with the customers and the many external providers of sourced IT services. With such a diverse range of customers, inevitably all would feel special and all would have special wishes.

> Some business units might want a Mercedes. But if the business need is a premium car, it is up to the IT function to go into a constructive dialog to help that business choose an Audi or BMW, where Coop has a fleet agreement, and understands enough about the real needs of that business partner to ascertain that their special needs are taken into account when the final choice of configuration of the Audi or BMW is decided. (Munk-Nissen 2013)

Another key feature of the IT organization is the 'Support and maintenance' within 'IT development.' Munk-Nissen felt that it was the responsibility of development to make sure that whatever was developed could be run in an utterly efficient and reliable fashion. Accordingly, development was not 'off-the-hook' until the new systems were fully implemented and had been in operation for a while. Hence, the vertical 'black bars' called 'Service transition.' Actually, these vertical black bars would be a very important feature of the new organization, as they indicate the relations/transitions/hand-overs between the major functions and phases in the IT value chain.

Looking at the organization's IT resource expenditure, previously around 80% had been spent on the back-end of the value chain. An essential part of the new strategy was to move the majority of the people into the front-end in order to get closer to the customer. In this way, COOP could better identify the right business needs to be served by IT, get the projects on track, execute them on specifications and in accordance with customers' needs, and have a more standardized and simplified back-end system, all of which would support effectiveness, agility, and economy-of-scale.

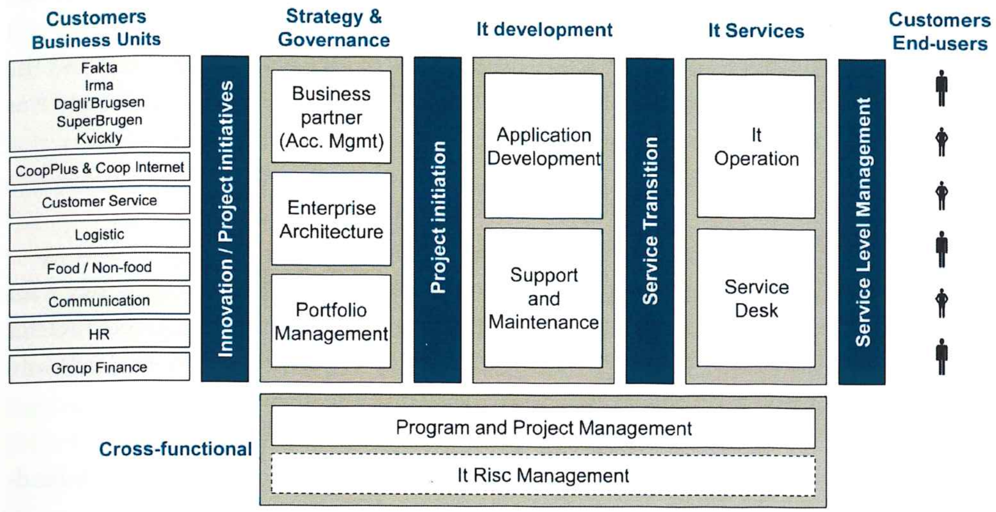

Figure 3. The new high-performing IT organization at Coop implemented in 2014

IT governance

With the new IT organization in place, it was time to face the huge challenge of the technological debt: the IT infrastructure had to be modernized, the mainframe had to be phased out, antiquated applications had to be retired, and state of the art applications had to be brought in.

This required a strong governance structure and a strong commitment from top management. Munk-Nissen had previously had the experience of launching a big IT governance program in which the directors were not very eager to join the different IT governance boards. In Coop, he promised to conduct a 360 degree evaluation to minimize the risk of, e.g., finding out – just before going live – that the system was not integrated with the supply chain. A model was made of how to prioritize and estimate value, based on how to add new value and protect existing value. These evaluations were based on some eight high-level criteria within key parameters concerning how easy it was to fulfill the project and what the generated value is.

Although several managers needed some persuasion, Munk-Nissen believed that the evaluation and the IT governance structure would ensure stable functionality and that milk would continue flowing to the shops. When it came to portfolio management and IT governance, he recognized the danger of short-term horizons and conflicting interests between the different business units within the company. However, with the introduction of the portfolio manager, strict evaluations would lay the foundation for an effective governance process. Projects would no longer just be rushed through. In this way, inch-by-inch, Munk-Nissen eventually had a full scale IT governance program in place.

In this way, Coop IT wanted not only to be the provider of IT governance and the facilitator of all IT sourcing, but it also wanted to ensure that its IT governance was supportive of the strategy of Coop and the major decision processes.

The new IT strategy

Realizing the challenges in early 2013, Munk-Nissen developed a new strategy for IT in Coop, labeled the 615 strategy. It got its name from the three strategic metrics characterizing the new IT strategy that would enable IT to meet business needs and expectations of Coop with readiness and delivery on time, every time:

- deliver six times the business value per dollar spent on operating the IT landscape; (the six)
- deliver a 100% stable IT operation; (the 100)
- by 2015 (the 15)

The ambition to deliver 6 times the value requires a little explanation. If current division of every DKK spent on IT is 80/20 on operations and on build, respectively, the ambition is to get to a situation of 40/60 for run/build. In the old case, for every DKK 1.00 spent on run, only DKK 0.25 was spent on build. In the dream scenario of 40/60, for every DKK 1.00, the amount spent on build in 50% higher, i.e., DKK 1.50. Accordingly, relatively six times more on build for every DKK spent on run! (Munk-Nissen 2013)

Moving forward with SAP

One of the first major chunks of technological debt to be addressed was the HR and Finance function. In 2012, the board decided to integrate the support functions of the different supermarket chains in order to achieve economy of scale. A highly publicized consequence prior to this was that the head quarter and administration of Irma was moved to Albertslund. Because of this, the CEO of Irma, Alfred Josephsen, decided to leave Coop.

The reason to start with HR and Finance was that this was a prerequisite for Coop to function as one integrated retail company. Staff should be recruited and utilized across all chains, and it should be possible to seamlessly consolidate the finances for the whole group. Another reason to start with HR and Finance was that it was considered a good place to start learning about how to implement such huge ERP modules

from SAP before a possible embarking on implementing SAP retail for the full supply chain.

Early on, it was decided not to develop own software. This did not mean, however, that Coop wanted to completely stop building systems themselves, but systems within HR and Finance were hardly core to a retail supermarket, and it seemed much more advantageous to avail oneself of a standard package offered by SAP under the motto of 'simplify, standardize, and consolidate.' The budget was DKK 250 million, which was a major investment, but it was considered necessary, and, at any rate, would be cheaper than developing new systems from scratch. Both systems were planned to go live on 1 January 2014.

In accordance with the governance model, one project group was appointed for each of the two business support functions having members both from IT and from the business side. The project chairman in each of the two groups came from the business side, i.e., from HR and Finance, respectively. There was also a program steering committee for these two projects including the program director, CFO and CIO (Munk-Nissen).

Even though it was standard ERP modules from SAP, which had been implemented in many retail organization systems, the implementation in Coop went far from smooth. In fact, the starting date of the Finance module was postponed several times, and both systems were not fully operational until 1st January 2016 when the total costs had substantially exceeded the budget.

Munk-Nissen gave one overarching explanation: Improper management and project execution. More precisely, he identified a number of critical issues:

- The identification of business needs had not been carried out thoroughly enough. The project groups made a number of assumptions about how the business operated, which were simply not aligned and verified across the 1200 shops.
- In the development, it was assumed that the systems were to support standard business processes, but these turned out to be far from standardized in the shops.
- A number of problems had not been foreseen and they remained unnoticed until far too late in the process, and the development team had to go back and modify the systems.
- A special challenge was the approximately 400 franchise shops, which were operating in many different and often more complicated ways.
- The franchise stores were also rather reluctant to change their processes for some higher 'standard' Coop cause, without a clear picture on how it would benefit them individually.
- The steering committee governance was challenged by the human fallacy of

preferring a "green light" regime. Management teams are typically not very happy with delays and budget overruns. Accordingly, also in this case, there was an in-built tendency for the program director to emphasize the good news and be a little more frugal with the bad news, until these were too difficult to contain.

In summary, the experiences with introducing the HR and Finance SAP modules were very stressful, and top management now had the impression that introducing the more comprehensive SAP retail business module covering the full supply chain would be too complex and too risky.

However, something had to be done about the technological debt: Needed was at least DKK 600 million to renew the infrastructure, DKK 200 million to renew the cashier terminals, DKK 200 million to rationalize the many servers and infrastructure in the shops (for price look-up, for cashier checkout, for store control, etc.), plus another three digit million investment to renew the core retail supply chain management. Whether it came from SAP or another ERP vendor probably did not matter that much. What did matter was that Coops biggest competitor Dansk Supermarked had modernized its infrastructure and completed their SAP implementation for the full supply chain management in 2015, and Coop was still running many different, up to 20-year-old systems, with high risks of failure and low possibilities of innovation.

However, how could they finance an investment of more than DKK 2 billion in IT infrastructure and applications? With the current profit margin of 1%, it would take the total profit margin (EBIT) for more than four years to achieve a state of the art IT shop.

Questions

1. How important is IT in the retail business? What would happen, if central systems were not available for hours or even days?
2. How do you assess the so-called 'high performance IT organization' put forward by Munk-Nissen? What do you see as the strengths and weaknesses?
3. How do you assess the 615 strategy proposed by Munk-Nissen?
4. How do you assess the severity of the technological debt of Coop?
5. What would you suggest that Coop should do?

Appendix 1